

Asian American and Pacific Islander Breast and Cervical Carcinoma Screening Rates and Healthy People 2000 Objectives

Marjorie Kagawa-Singer, Ph.D., M.N., R.N.¹
Nadereh Pourat, Ph.D.²

¹ Department of Community Health Sciences and Asian American Studies, UCLA School of Public Health, Los Angeles, California.

² UCLA Center for Health Policy Research, Department of Health Services, UCLA School of Public Health, Los Angeles, California.

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Address for reprints: Marjorie Kagawa-Singer, Ph.D., M.N., R.N., Department of Community Health Sciences and Asian American Studies, UCLA School of Public Health, 650 Charles E. Young Drive South, CHS 26-051, Los Angeles, CA 90095-1772.

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BACKGROUND. Breast carcinoma is the number one cause of cancer-related mortality among Asian American and Pacific Islander (AAPI) women, and for certain subgroups cervical carcinoma is among the top three causes. Yet AAPI women have the lowest cancer screening and early detection rates of all ethnic groups. The practice of merging AAPI data hides significant variation among AAPI groups, and the groups at highest risk for late presentation of disease become invisible. To the authors' knowledge, this study is the first report of disaggregated national data for breast and cervical carcinoma screening in AAPI subpopulations. The authors used data from the National Health Interview Survey (NHIS 1993 and 1994) and evaluated major socioeconomic and access variables associated with screening practices.

METHODS. Healthy People 2000 (HP2000) criteria were used to evaluate the most recent NHIS screening rates for AAPIs as one group and for six distinct subgroups compared with HP2000 criteria. The sample included 2756 non-Hispanic AAPI and 64,196 non-Hispanic white women age 18 years or older in the 1993 and 1994 NHIS. The effects of socioeconomic and access-to-care indicators on screening were explored.

RESULTS. Breast and cervical carcinoma screening rates for AAPIs were below those for white women and well below national goals and guidelines. Lack of insurance, low income, and lack of a usual source of care had a significantly greater negative impact on AAPIs than on white women, indicating that AAPIs may be more vulnerable to structural barriers to cancer control efforts.

CONCLUSIONS. AAPI women have very low rates of breast and cervical screening, which increases their chances of later stage disease presentation. To reach the Healthy People 2000 objectives, targeted efforts are needed to increase cancer screening among AAPI women overall and specific subgroups in particular. *Cancer* 2000;89:696-705. © 2000 American Cancer Society.

KEYWORDS: breast carcinoma screening, cervical carcinoma screening, Asian Americans, women, cancer screening guidelines.

Cancer is the number one cause of death among Asian American and Pacific Islander (AAPI) women,¹ and, at the same time, AAPIs have the lowest screening rates of all ethnic groups in the United States.² Despite these facts, AAPIs, the fastest growing population in the United States (7.5 million in the 1990 Census), have received the least attention of all ethnic populations in cancer control research studies or targeted intervention programs by the national government.³ One of the major reasons for this lack of attention is the paucity of disaggregated and accurate AAPI morbidity, mortality, and

screening data.¹ Aggregated data imply a lack of need for targeted screening programs and public policies for these populations.

AAPIs themselves, as well as policy makers, labor under the myth that AAPIs are a healthy population and are at low risk for breast and cervical carcinoma. Compared with whites, however, AAPI women present later with more advanced disease. Breast carcinoma in these women appears at a younger age and is of a more aggressive type than that seen in other ethnic groups.⁴ For cervical carcinoma, Southeast Asian (SEA) women present much later and have the highest mortality rate of all ethnic groups.⁵ To our knowledge, no national studies have been conducted on the cancer screening rates of subgroups of AAPI women from national data bases. This paper is the first report of national breast and cervical carcinoma screening rates for disaggregated groups of AAPI women.

BACKGROUND

The erroneous notion that AAPIs are at low risk for developing cancer is fostered by assuming incorrectly that all AAPIs belong to a homogeneous culture. The result is that a dichotomous ethnicity variable (AAPI or not AAPI) is used in research as a proxy for culture and is reported to be a risk factor for a health behavior: a common error in most studies.^{6,7} AAPIs are the only ethnic population in the United States with a bimodal distribution of major demographic risk factors: education, income, insurance status, and age.

The practice of aggregating over 60 nationalities into one AAPI category masks their marked differences in major demographic characteristics that are known to be associated with risk factors for low cancer screening. For example, the average college education rate for Chinese Americans is 42% compared with the U.S. average of 21%; however, the number of women with less than 4 years of education was 2% for the U.S. and 6% for AAPIs. No subgroup breakdown is available for AAPIs with less than 4 years of education.⁸ The Japanese American poverty rate is 6.6%, or the same as whites, whereas the rate for all SEA groups is over 66%.⁸ The median income for AAPIs for a family of four in 1993 was \$44,460; however, for Asian Indians, the median income was \$60,903, and, for the Hmong, the average income was \$20,648.⁹ Although the AAPI median aggregated age is slightly lower than the median U.S. aggregated age (31.2 years vs. 33.4 years), the Vietnamese and Hmong median ages are 25.2 years and 12.5 years, respectively. Certain subgroups, such as Korean and Vietnamese elderly, have a higher proportion of uninsured and poorer self-reported health status than any other ethnic group.¹⁰

Apart from demographics, AAPIs vary greatly in

acculturation and English fluency levels, because > 65% are foreign born. A large proportion of AAPIs is isolated linguistically (32.8%), with significant variation among the subgroups: Hmong, 55.8%; Chinese, 40.8%; Filipino, 17%; and Samoan, 14.2%.^{8,11} Non-English speaking AAPIs usually are excluded from national surveys, including the National Health Interview Survey (NHIS), and they tend to have the lowest screening rates.

Although breast and cervical carcinoma rates in the native Asian and Pacific Islander countries are 25–50% of those of women in the United States population, acculturation studies indicate that cancer rates for AAPIs increase and begin to mirror the rates of the host country within 10 years of immigration.¹² The current average age of AAPIs is significantly lower than white Americans, but the combination of increasing age and acculturation portends major increases in the incidence of the most common cancers for AAPIs, as exemplified in American Cancer Society, California Division statistics (1999). For example, the cancer rate for Japanese Americans, the most acculturated AAPI subgroup with the oldest population, is more than three times as high as Koreans, a more recent and younger immigrant group. To address the shortcomings of current research on AAPI health behavior and use, we disaggregated the AAPI groups, given the limitations of the data, into six groups and explored the differences in screening practices by income and insurance coverage to account for within group differences.

Cancer Incidence

AAPI incidence and mortality rates for breast and cervical carcinoma are lower overall than their white counterparts until this population is disaggregated, and significant differences in incidence rates emerge.⁵ Between 1988 and 1992, AAPI subgroup breast carcinoma incidence rates varied from 74.0 for Japanese to 23.3 per 100,000 for Koreans.¹³

The variation among AAPI subgroups also is reflected in the ranking of the sites of their tumors. For example, the uterine cervix is the second most common site for tumors in Vietnamese women (43.8%) as well as in other SEA women (35.7%) but not in Asian-Indian, Chinese, or Japanese women. Disease stage at diagnosis also varies: Japanese, Vietnamese, Filipino, and Korean women all are diagnosed with cervical carcinoma at *later* stages than white women. No data are available for most AAPI subgroups.

Screening Practices and Objectives

AAPIs have the lowest screening rates of all ethnic groups. Even though survival rates for breast and cer-

vical carcinoma are excellent with early detection (85–99%),¹ and consistent screening improves the likelihood of in situ stage detection, AAPI cancer deaths have increased 200% in the last 10 years compared with a 30% increase among other ethnic groups.¹

Healthy People 2000 (HP2000) screening goals indicate that 95% of women age > 18 years should be screened annually for cervical carcinoma. The guidelines also established mammography rates at 80% for all women age $\geq$ 40 years who ever received this examination and 60% for women age > 50 years who received a mammogram examination in the last 1–2 years. Women ages 40–49 years should be screened biannually unless they are at risk for breast carcinoma, and women age > 50 years should be screened annually.¹²

The degree to which these objectives are achieved among AAPI women nationally is unknown. Therefore, we posed the following two research questions: 1) What are the differences in rates of cancer screening for AAPI women and their subgroups compared with whites, and 2) are the differences in screening rates among AAPI and white women associated with differences in demographics, access, or health insurance coverage?

MATERIALS AND METHODS

Data and Sample

We used data from the NHIS, a stratified, multistage survey of the civilian population of the United States that is repeated each year with a new independent sample. We used the most recent data available, 1993 and 1994, to obtain the maximum possible sample size and to increase the reliability of the estimates for subgroups of AAPI women (the more recent 1995 and 1996 NHIS data do not include cancer screening questions). The total sample of non-Latino white women age $\geq$ 18 years was 64,196, and it was 2756 for AAPI women.

We obtained information on demographic variables and several health care utilization measures from the core survey. Information on health insurance coverage and the sources of this coverage came from the health insurance supplement. Data on cancer screening were obtained from the HP2000 supplement. In 1993, the HP2000 supplement was administered in the second half of the year and, thus, has half the sample size of the larger core survey. Consequently, the sample for the analysis of screening practices is limited to the complete sample for 1994 and half of the sample for 1993. This limitation led to a total of 17,373 non-Latino white women ages $\geq$ 18 years and 528 AAPI women for the study of cancer screening.

Similar to other national surveys, the NHIS is conducted in English and Spanish only. Interviewers attempt to interview those who speak other languages using a translator within the household. Therefore, the findings of this study are limited to AAPI women who are not linguistically isolated.

We adjusted the weights provided by NHIS to account for combining the full-year data for 1994 and half-year data for 1993 and to represent the population of AAPI women in the U.S. for the 1993–1994 period. Based on NHIS recommendations, the 1994 sample was given a two-thirds weight, and the 1993 half-year sample was given a one-third weight. We used the SUDAAN software statistical package (release 7.0; Research Triangle Institute, Research Triangle Park, NC)¹⁴ to calculate the standard errors of all estimates, because the NHIS has a stratified, multistage design that results in artificially reduced variation in the sample. All of the standard errors presented in Tables 1–4 are corrected for design effect by SUDAAN.

Variables

We compared Papanicolaou (Pap tests) and mammogram screening rates of the non-Latino whites with the following subgroups of non-Latino AAPIs: Chinese, Filipino, Korean, Vietnamese, Japanese, and Asian Indians. Due to insufficient sample sizes, the data on Hawaiians and other Pacific Islanders could not be analyzed and are not reported. Rates of Pap tests were studied for women ages $\geq$ 18 years, and mammogram rates were studied for those ages $\geq$ 40 years in accordance with the recommendations of HP2000.¹²

We performed confirmatory analysis of demographic differences between AAPI subgroups and further analyzed the effects of poverty level and educational levels on screening outcomes, because both are strongly related to under-use of preventive health care.^{15,16} The poverty level in NHIS is based on the U.S. Department of Commerce, Bureau of the Census poverty thresholds, in which poverty is calculated as the proportion of income per number of people in the family. The poverty level is then calculated by comparing the person's income with the poverty thresholds. We categorized the poverty level into 0–99%, 100–199%, 200–299%, and 300% or more of the Federal Poverty Level (FPL). Education was categorized as 0–5 years, 6–12 years, and $\geq$ 13 years of education.

Having health insurance coverage is related positively to receiving preventive services.^{17,18} Having insurance is a proxy for ability to afford, have access to, and use care.

In addition, we explored the effects of three access indicators on screening services. Comparisons were

TABLE 1
Sociodemographic Characteristics of Women Age ≥ 18 Years by Ethnicity, 1994–1995^a

Characteristic	White	Asian American Pacific Islanders						Asian Indians
		Total AAPIs	Chinese	Japanese	Filipino	Korean	Vietnamese	
Sample size	64,196	2756	648	323	618	218	257	320
Age (yrs)								
18–39	42 (0.3)	54 (1.5) ^b	54 (1.8) ^b	36 (2.5) ^d	53 (1.8) ^b	55 (3.3) ^b	60 (2.6) ^b	62 (2.2) ^b
40–49	19 (0.2)	21 (0.6) ^b	23 (1.6) ^d	20 (1.2)	22 (1.8)	20 (2.6)	20 (1.9)	21 (2.0)
50–64	19 (0.2)	16 (0.9) ^b	15 (0.2) ^c	23 (2.3)	15 (1.6) ^d	16 (2.3)	16 (1.8)	12 (1.6) ^b
> 65	21 (0.3)	9 (1.1) ^b	8 (0.7) ^b	21 (3.3)	9 (1.3) ^b	9 (2.4) ^b	5 (1.2) ^b	4 (1.0) ^b
Poverty level (%)								
< 100	9 (0.3)	16 (1.0) ^b	13 (2.2)	7 (1.8)	9 (1.8)	13 (2.7)	36 (4.0) ^b	15 (2.5) ^d
100–199	23 (3.2)	25 (1.2)	28 (1.9) ^c	16 (2.3) ^c	28 (3.1)	24 (2.6)	26 (3.7)	21 (2.9)
200–299	21 (0.2)	20 (1.0)	17 (2.4)	16 (1.9) ^c	23 (1.8)	25 (3.5)	16 (3.7)	19 (2.8)
> 300	46 (0.5)	40 (1.4) ^b	42 (2.6)	60 (2.8) ^b	40 (2.8) ^d	38 (3.4) ^d	22 (4.0) ^b	44 (3.6)
Education (yrs)								
0–5	1 (0.1)	6 (0.6) ^b	6 (0.9) ^b	—	2 (0.6)	5 (2.0) ^d	14 (2.1) ^b	4 (1.1) ^c
6–12	0.6 (4.7)	0.42 (1.6) ^b	0.42 (2.0) ^b	0.45 (3.2) ^c	0.31 (1.9) ^b	0.48 (3.6) ^b	0.6 (4.3)	0.32 (3.0) ^b
13–18	43 (0.5)	52 (1.4) ^b	52 (2.1) ^b	55 (3.2) ^c	66 (1.8) ^b	47 (2.9)	31 (3.4) ^c	65 (3.2) ^b
Yrs in United States								
< 10	1 (0.1)	34 (2.6) ^b	41 (2.9) ^b	12 (5.1) ^d	34 (3.0) ^b	30 (3.5) ^b	48 (3.9) ^b	50 (3.9) ^b
> 10	4 (0.1)	45 (2.9) ^b	42 (2.3) ^b	29 (6.5) ^c	53 (2.1) ^b	61 (4.0) ^b	49 (4.1) ^b	45 (4.0) ^b
Born in U.S.	95 (0.1)	21 (5.2) ^b	17 (2.4) ^b	59 (11.1) ^c	14 (2.7) ^b	9 (2.1) ^b	2 (0.8) ^b	5 (1.3) ^b
Health insurance coverage								
Uninsured	12 (0.2)	20 (2.1) ^c	25 (2.4) ^b	9 (3.5)	17 (3.0)	31 (4.3) ^b	25 (3.6) ^c	22 (4.3) ^d
Other	84 (0.3)	71 (2.5) ^b	70 (2.5) ^b	89 (3.4)	77 (2.6) ^c	65 (4.6) ^b	50 (4.6) ^b	74 (4.1) ^d
Medicaid	4 (0.2)	8 (1.2) ^c	6 (1.1)	—	6 (1.4)	5 (1.5)	25 (4.6) ^b	4 (1.1)

^a The probability values represent the comparison of Asian American Pacific Islanders (AAPIs) or their subgroups with whites. All standard errors (in parentheses) were adjusted for design effect with SUDAAN. Cell sizes less than 5 are not reported.

^b $P < 0.0001$.

^c $P < 0.01$.

^d $P < 0.05$.

made between the rates of cancer screening for those who had a doctor visit in the past year and those who did not, for those who had a regular source of care and those who did not, and for those whose regular source of care was a private doctor's office or health maintenance organization (HMO) as opposed to a public clinic or a hospital. All three access indicators have been shown to increase the likelihood of use of preventive care.^{19,20} A woman has a higher likelihood of being offered a Pap test or a mammogram if she visits a health care provider than if no visit took place. Similarly, the chance of being offered either one of these screenings is higher with a regular source of care or when the source of care is a private doctor or HMO.

The subgroups presented in this article are those with a minimum cell size of 6 to support the analysis. Because of this, we could not analyze the contribution of demographic or access indicators for groups such as Vietnamese, who are the most disadvantaged by socioeconomic status measures and are more likely to have lower screening practices. The subsets of Chi-

nese, Filipino, and Japanese women more often had adequate sample size and, thus, are reported more consistently. We compared the differences in screening rates using probability levels as high as 0.05 to assess the significance of differences found. The tables identify significant differences between whites and AAPIs. Where appropriate, we compared differences among AAPI subgroups or within a group, but these differences are reported in the text only to avoid confusion in comparisons within the tables.

RESULTS

Demographic Profile of AAPI

Table 1 presents the demographic profile of the sample of AAPI women ages ≥ 18 years ($n = 2756$ women). About 3.4% of the NHIS population of adult women ages ≥ 18 years identified themselves as AAPI, which is comparable to their representation in the United States (3.7%).²¹ The five largest ethnic AAPI groups, ages ≥ 18 years, were Chinese (24%; $n = 648$ women), Filipino (22%; $n = 618$ women), Japanese (12%; n

TABLE 2
Percentage of Papanicolaou Smear Screening for Women Ages > 18 Years and Mammograms for Women Ages > 40 Years by Ethnicity, 1994–1995^a

Characteristic	White	Asian American Pacific Islanders						
		Total AAPIs	Chinese	Japanese	Filipino	Korean	Vietnamese	Asian Indians
Time since Pap smear								
Sample size	17,373	528	128	72	123	50	35	60
Past yr	51 (5.5)	43 (2.7) ^c	42 (4.6)	45 (4.2)	47 (5.0)	33 (9.2) ^d	25 (7.3) ^c	43 (6.9)
1–3 yrs ago	26 (4.3)	24 (2.1)	16 (3.8) ^c	34 (4.2)	27 (5.1)	37 (8.5)	30 (10.3)	20 (5.8)
> 3 yrs	18 (0.4)	12 (1.6) ^c	13 (3.3)	13 (3.0)	10 (3.9) ^d	—	—	—
Never had one	5 (0.3)	21 (2.3) ^b	28 (5.0) ^b	8 (2.9)	15 (4.6)	25 (8.4) ^d	36 (11.9) ^c	26 (6.1) ^c
Time since mammogram								
Sample size	10,567	213	54	48	44	22	11	15
Past yr	41 (0.6)	37 (3.3)	41 (7.4)	43 (4.9)	43 (10.1)	39 (14.2)	—	—
1–2 yrs ago	20 (0.5)	19 (2.3)	23 (22.7)	20 (3.7)	16 (5.4)	—	—	—
More than 2 yrs	18 (0.5)	14 (3.1)	—	17 (4.1)	13 (6.3)	—	—	—
Never had one	21 (0.6)	30 (4.1) ^d	26 (8.8)	21 (4.5)	28 (8.7)	50 (15.2)	—	68 (11.6) ^b

^a The probability values represent the comparison of Asian American Pacific Islanders (AAPIs) or their subgroups with whites. All standard errors (in parentheses) are adjusted for design effect with SUDAAN. Cell sizes less than 5 are not reported.

^b $P < 0.0001$.

^c $P < 0.01$.

^d $P < 0.05$.

= 323 women), Asian Indian (12%; $n = 320$ women), and Vietnamese (9%; $n = 257$ women), essentially the same as the distribution in the United States of the top five AAPI groups.

Adult AAPI women generally are younger than white women, with more than half (54%) between the ages of 18–39 years, significantly more than whites (42%). Within AAPI women, Japanese had the lowest percentage of young women (36% were ages 18–39 years), partly as the result of lower recent immigration rates.⁹

Many AAPI women live in poverty or near poverty, with significant variations within subgroups. A higher percentage of AAPI women (16%) than white women (9%) are below 100% of the FPL, with low rates for Asian Indian women (15%) and high rates (36%) for Vietnamese women.

A distinct bimodal distribution exists for levels of education of AAPI women. More AAPI women than white women have less than 6 years of education (6% vs. 1%), but more AAPI women than white women have a college education (52% vs. 43%). Certain groups, such as Asian Indians (65%) and Filipinos (66%), have the highest rates of college education of all ethnic groups.²¹

The majority of the AAPI women are immigrants to the United States (79%). The highest recent immigration rates (<10 years) were among Asian Indians (50%) and Vietnamese (48%).

AAPI women were significantly more likely to be uninsured compared with white women (20% vs. 12%, respectively), with the highest rate for Korean women (31%). Twice as many AAPI women have Medicaid, the public source of health insurance coverage for low-income individuals (8% vs. 4%), and Vietnamese have the highest rate of all AAPI women at 25%.

Cancer Screening Rates of AAPI Women

The data in Table 2 address research question 1: What are the differences in rates of cancer screening for the AAPI women and their subgroups compared with white women? The data indicate that the cancer screening rates for AAPI groups are well below the HP2000 guidelines and below white women in almost every category.

HP2000 recommends that 95% of women age ≥ 18 years should receive a Pap test. We found that one-fifth (21%) of AAPI women have never had a Pap test compared with only 5% of white women. Among AAPI women, Japanese women (8%) do not differ significantly from white women (5%), but significantly more Vietnamese (36%), Chinese (28%), Asian Indian (26%), and Korean (25%) women have never had a Pap test.

AAPI women are screened at longer intervals than are recommended in HP2000. Approximately 1 of every 7 AAPI women (12%) has not received a Pap test for 3 or more years, a period that can render this screening ineffective. Even among those who receive

TABLE 3
Percentage of Women Who Never Had a Papanicolaou Smear Screening Ages > 18 Years, by Sociodemographic Characteristics and Access, 1994–1995^a

Characteristic	White	Asian American Pacific Islanders				
		Total AAPIs	Chinese	Filipino	Korean	Asian Indians
Education (yrs)						
0–5	15 (3.1)	—	—	—	—	—
6–12	5 (0.3)	25 (3.6) ^b	34 (8.5) ^c	26 (10.1) ^d	14 (8.3)	51 (12.0) ^c
13–18	4 (0.4)	18 (2.7) ^b	25 (6.6) ^c	11 (5.3)	33 (12.4) ^d	18 (5.9)
Poverty level (%)						
< 100	11 (1.4)	48 (8.1) ^b	53 (13.7) ^c	—	58 (14.2) ^c	72 (13.6) ^b
100–199	6 (0.5)	21 (0.5) ^c	26 (9.9) ^d	25 (11.7)	—	—
200–299	4 (0.4)	22 (5.1) ^c	43 (9.4) ^b	—	—	—
> 300	3 (0.3)	8 (2.4) ^d	12 (4.5) ^d	—	—	—
Insurance						
Insured	5 (0.3)	14 (2.1) ^b	22 (5.6) ^c	9 (9.3)	23 (7.9) ^d	15 (5.5)
Uninsured	7 (0.7)	48 (5.3) ^b	51 (13.8) ^c	46 (15.9) ^d	—	65 (11.8) ^b
Visited the doctor last year						
Yes	4 (0.3)	15 (2.0) ^b	18 (4.0) ^c	15 (4.9) ^d	24 (8.4) ^d	19 (6.1) ^d
No	8 (0.7)	37 (4.9) ^b	56 (10.2) ^b	—	24 (12.6)	35 (13.0) ^d
Has a regular source of care						
Yes	0.04 (0.3)	14 (2.3) ^b	19 (5.2) ^c	11 (4.3)	25 (9.3) ^d	11 (4.5)
No	9 (0.9)	50 (6.7) ^b	62 (9.4) ^b	49 (0.2) ^d	—	66 (12.9) ^b
Type of regular source of care						
Private or HMO	4 (0.2)	11 (2.9) ^d	16 (5.4) ^d	12 (4.9)	23 (10.1)	—
Public or other	8 (1.5)	24 (10.2)	49 (16.9) ^d	—	—	—

^a The probability values represent the comparison of Asian American Pacific Islanders (AAPIs) or their subgroups with whites. No sample sizes are presented, because each cell had a different sample size. Instead, standard errors (in parentheses) are used to indicate reliability, so that, when they are greater than 30% of the estimate, the data are considered unreliable. All standard errors are adjusted for design effect with SUDAAN. Cell sizes less than 5 are not reported.

^b $P < 0.0001$.

^c $P < 0.01$.

^d $P < 0.05$.

timely Pap tests, AAPI women lag behind white women. Fifty-one percent of white women received Pap tests in the year previous to the survey, whereas 43% of AAPI women did so.

Despite the HP2000 recommendation that all women age > 40 years should be screened at least biannually (1991), more AAPI women (30%) have never had a mammogram compared with white women (21%). The percentage of Japanese women who never had a mammogram is similar to that for white women, but Asian Indian women (68%) have a far higher percentage of no past mammograms compared with Japanese women ($P < 0.001$).

Tables 3 and 4 address research question 2: Are the differences in screening rates among AAPI women and white women explained by the differences in demographics, access, or health insurance coverage? We selected women who never had a Pap test or mammography and compared the rates of insurance coverage, office visits, and other access indicators among white women and AAPI groups of women. We found

significant differences in screening rates within different categories of socioeconomic factors for AAPI women compared with white women.

Education

For Pap testing (Table 3), education is a significant predictor of screening for white women. More uneducated white women or those with an elementary education (15%) never had a Pap test compared with white women with a high school or college education (5% and 4%, respectively).

Other research has shown that more AAPI women had little or no education compared with white women (6% compared with 2% in the U.S.).²² We were unable to detect the differences in screening rates between AAPI women and white women with little or no education, because the NHIS was implemented only in English, and those AAPI women with a limited education are more likely to be monolingual and, thus, are less likely to have participated in the survey. Nevertheless, we found that AAPI women with high

TABLE 4
Percentage of Women Who Never Had a Mammogram Screening, Ages > 40 Years, by Sociodemographic Characteristics and Access, 1994–1995^a

Characteristic	White	Asian American Pacific Islanders					Asian Indians
		Total AAPIs	Chinese	Japanese	Filipino	Korean	
Education (yrs)							
0–5	39 (4.5)	—	—	—	—	—	—
6–12	25 (0.7)	33 (7.36)	40 (14.20)	—	53 (23.14)	43 (19.49)	—
13–18	14 (0.7)	25 (4.90) ^d	15 (8.26)	30 (5.94) ^c	20 (7.96)	—	—
Poverty level (%)							
< 100	38 (1.9)	64 (9.75)	—	—	—	—	—
100–199	32 (1.1)	25 (6.81)	—	—	—	—	—
200–299	21 (1.2)	36 (8.95)	—	—	32 (13.95)	—	—
> 300	14 (0.6)	22 (4.17)	—	—	—	—	—
Insurance							
Insured	20 (0.6)	24 (3.73)	14 (7.09)	18 (6.09)	24 (7.56)	34 (14.61)	48 (14.73)
Uninsured	42 (2.8)	72 (8.36) ^c	—	—	—	—	—
Visited the doctor last year							
Yes	18 (0.5)	19 (0.51)	15 (9.04)	17 (3.97)	27 (10.23)	33 (13.88)	58 (13.12) ^c
No	43 (1.5)	43 (1.32)	63 (14.40)	—	37 (15.82)	—	—
Has a regular source of care							
Yes	19 (0.6)	24 (4.03)	18 (8.33)	15 (3.41)	28 (8.86)	28 (14.43)	55 (14.19) ^d
No	46 (2.2)	73 (9.04) ^c	76 (15.37) ^d	—	—	82 (13.68) ^c	—
Type of regular source of care							
Private or HMO	19 (0.6)	25 (4.16)	19 (8.62)	15 (3.28)	31 (9.28)	—	56 (15.61) ^d
Public or other	24 (2.7)	12 (5.74) ^d	—	—	—	—	—

^a The probability values represent the comparison of Asian American Pacific Islanders (APIs) or their subgroups with whites. No sample sizes are presented, because each cell has a different sample size. Instead, standard errors (in parentheses) are used to indicate reliability, so that, when they are greater than 30% of the estimate, the data are considered unreliable. All standard errors are adjusted for design effect with SUDAAN. Cell sizes less than 5 are not reported.

^b $P < 0.0001$.

^c $P < 0.01$.

^d $P < 0.05$.

school and college educations varied in their Pap rates. Among AAPI women with a high school education, many Asian Indian women (51%) had never had a Pap test, and, among those women with a college education, the fewest (11% of Filipinos and 18% of Asian Indians) remained untested.

Education has a similar effect on mammography screening for white women (Table 4), with the additional significant difference between high school-educated and college-educated white women (25% vs. 14%). However, AAPI high school-educated and college-educated women do not differ significantly in their mammogram rates, most likely due to small sample sizes and high variation.

Poverty

Poverty exerts a significant influence on Pap smear screening rates for both ethnic groups, but it is more pronounced for AAPI women. Over 1 in 10 (11%) white women below 100% of the FPL never had a test compared with 3% of those at 300% or more of the FPL. AAPI women's rates were 48% and 8%, respectively. Compar-

ing the rates of no previous Pap test of white women and AAPI women within different poverty groups shows discrepancies in screening rates, and these rates are more apparent among lower income groups. For example, among those below 100% of poverty, the rate of no past Pap test in white women is 11%, and, in AAPI women, it is 48%, with Asian Indian rates as high as 72%. Among those at 100–199% of the FPL, the rate of never having had a Pap test in white women is 6%, and, in AAPI women, it is 21%. Even among the highest income groups, the difference between white women and AAPI women remains significant.

The effect of poverty on the prevalence of mammography screening is somewhat similar to that of Pap smears. For both white women and AAPI women, as poverty decreases, the rates of women who had no previous mammograms decrease. However, significant differences in mammogram rates remain. For those below 100% of the FPL, 38% of white women, compared with 64% of AAPI women, never had a mammogram.

Effect of Health Insurance, Use, and Access

Lack of health insurance seems to have a greater effect on screening for AAPI women than for white women. Forty-eight percent of uninsured AAPI women never had a Pap test compared with only 7% of uninsured white women ($P < 0.001$). The same substantial difference is observed for AAPI subgroups. In comparing the groups, insurance has a relatively negligible effect on Pap test screening for white women. Whereas the disparity between the uninsured AAPI women and white women is 41% points ($P < 0.001$), the disparity between the insured AAPI women and white women is only 9% ($P < 0.001$).

Insurance has an even greater effect on the prevalence of mammography screening. The proportion of insured AAPI women (24%) who never had a mammogram is dramatically lower than that of uninsured AAPI women (72%; $P < 0.001$) compared with 20% of insured and 42% of uninsured white women ($P < 0.001$). The gap between those who had no previous mammograms among the insured AAPI and white women (24% vs. 20%, respectively) is not significant, whereas the gap between the uninsured women is significant (72% vs. 42%, respectively).

Lack of access to or use of primary care also affects screening rates of AAPI women and white women. Thirty-seven percent of AAPI women who did not visit a doctor never had a Pap test (37%) compared with 15% of AAPI women. The effect of office visits on the rate of Pap tests was considerably smaller for white women than for AAPI women. The difference in Pap testing for white women without an office visit last year and those with an office visit was 4% (8% vs. 4%; $P < 0.001$) compared with 22% (37% vs. 15%) for AAPI women ($P < 0.001$). The effect of office visits on Pap testing shows a relatively consistent pattern among AAPI subgroups: Thirty-eight percent more Chinese women (18% vs. 56%) and 16% more Asian Indian women (19% vs. 35%) who had an office visit had a Pap test compared with those who did not visit a doctor last year.

Among those who did not have an office visit, AAPI women had a much higher rate (37%) with no past Pap tests compared with white women (8%). This substantial discrepancy points to the stronger effect of doctor visits on Pap screening for AAPI women compared with white women. Within AAPI women, the Chinese have the highest rates of no Pap screening (56%) compared with white women.

The magnitude of the effect on mammogram screening of having had an office visit is similar for AAPI women and white women. Forty-three percent of white women and AAPI women with no office visits

have never had a mammogram, and about 19% with office visits did not have a mammogram (19% for AAPI women, 18% for white women).

Having a regular source of care significantly increases the probability of both types of cancer screening. For example, the rate of no past Pap tests was higher for AAPI women without a regular source of care (50%; $P < 0.001$) compared with AAPI women with a regular source of care (14%; $P < 0.001$). Only 11% of Asian Indian women with a regular source of care did not have a Pap test compared with 66% of those without a regular source of care ($P < 0.001$). For Chinese and Filipino women, the effect was significant but was of a slightly smaller magnitude (19% and 62%, respectively for Chinese women, $P < 0.001$; and 11% and 49%, respectively, for Filipino women, $P < 0.05$). Of those without a regular source of care, significantly more AAPI women have not had a Pap test (50%) than white women (9%).

Having a regular source of care had an even greater effect on mammography screening. Among white women with a regular source of care, 19% never had a mammography compared with 46% of those with no regular source of care ($P < 0.001$). This difference is even more pronounced among AAPI women. About one-fourth (24%) of AAPI women with a regular source of care had no mammography compared with 73% of those without a regular source of care ($P < 0.001$).

Differences in the site of care or the type of provider reported as the usual source of care reveal an important effect on Pap screening of white women: Four percent of those with a private source of care, versus 8% of those with a public source of care, never had a Pap test ($P < 0.01$). However, the 13% difference between AAPI women with private and public sources of care is not significant. Comparing AAPI and white women with an HMO or a private doctor as their usual source of care, more AAPI women never had a test (11%) than whites (4%; $P < 0.01$).

Notably, the reverse correlation is observed for mammography rates of AAPI women when public care is compared with private sources of care. AAPI women with a public source of care have a lower rate (12%) of a past mammography screening compared with AAPI women with a private source of care (25%; $P < 0.01$). This discrepancy cannot be explained with the available data or analysis and may be due to the small sample sizes of AAPI women with public sources of care. White women have an expected pattern for their mammography: Nineteen percent with a private source of care never had a mammography screening compared with 24% of those with public sources ($P < 0.001$). Comparing white women and AAPI women

with public sources of care reveals a significantly higher rate of white women with no previous mammography tests (12% vs. 24%, respectively; $P < 0.01$).

CONCLUSIONS

This study confirms that 1) AAPI subgroups vary greatly in demographics and health insurance; 2) in aggregate, AAPI women have significantly lower rates of breast and cervical carcinoma screening than those recommended by HP2000 and than those of white women, and the same is even more pronounced in many AAPI subgroups; and 3) even with higher income, more education, insurance coverage, and better access to care, AAPI women and their subgroups still have lower rates of both cervical and breast carcinoma screening. These low rates of breast and cervical carcinoma screening reduce the chances of early detection and put AAPI women at particular risk for later stage presentation of these diseases.

Two major barriers to more in-depth knowledge of cancer prevention among AAPI women are the lack of sufficient and representative data^{22,23} and administration of national surveys in English only. Insufficient sample sizes hindered detailed analysis for most subgroups of AAPI women in this study. To enable more definitive analysis and to establish causal relations for all of the subgroups, (e.g., multivariate analyses), larger sample sizes are sine qua non. Without such data and analyses, specific predictors or risk factors cannot be identified. Over-sampling of the various subgroups is essential if we are to know more about this fastest growing ethnic group in the United States.

Second, the generalizability of such findings is limited due to the language limitations of existing national surveys. The women represented in the NHIS were either English speaking or had an English-speaking member of their household to assist them in answering the survey. Thus, about one-third of AAPI women who are monolingual are not represented in this survey. The monolingual segment of the AAPI women comprise an even higher risk group for late stage presentation of cancer because of language and access barriers.²⁴⁻³²

The bias created by the monolingual nature of the NHIS leads to an underestimation of the low rates of cancer screening among AAPIs. Between 1990 and 1997, at least 13 studies of AAPI breast carcinoma (9 studies) and cervical carcinoma (4 studies) screening have been conducted using population-based random samples and bicultural, bilingual interviewers to reach monolingual women.^{27,32-38} Eight surveys were conducted in California, and 1 was conducted in Texas. The groups surveyed were Vietnamese (5 studies), Korean (2 studies), and Chinese (2 studies).

Two other cervical carcinoma screening studies conducted between 1996 and 1998 in Seattle and Houston used convenience sampling methods to survey Cambodian American women, and two studies in San Francisco surveyed Vietnamese American women in their native languages. The rates for "ever having had" cervical carcinoma screening ranged from 25.5% to 76% for the Cambodian women and from 24% to 53% for the Vietnamese women. The range in the percentage of women age > 40 years who had ever had a mammogram was 17-73%, for Vietnamese women, 34-55% for Korean women, and 32-70% for Chinese women.

The majority of these studies were based on the Behavioral Risk Factor Surveillance Survey questions and indicated that, when monolingual and low-income AAPI women are included, the screening rates are even lower than the rates reflected in national population-based studies. These monolingual women generally are excluded in national studies due in part to language limitations.

AAPIs will constitute 10% of the population of the United States by 2050. Increasing numbers, acculturation, and aging all portend rapidly rising breast and cervical carcinoma rates. Policies and programs are needed to specifically target AAPI women for prevention programs to increase their rates of breast and cervical carcinoma screening. However, accurate disaggregated information from representative national surveys is essential to identify salient issues to promote earlier cancer screening and detection in this population. The NHIS and other national survey programs must collect data on the subgroups of AAPIs to identify baseline screening practices and to characterize structural and cultural barriers to screening for each group. Thus, effective programs can be implemented to promote earlier cancer screening and detection. Such programs will reduce the human and social costs of cancer in the AAPI population and reduce the discrepancy in health outcomes between whites and AAPIs. Targeted, cost-effective programming can be achieved only with accurate data.

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